

MACHINE LEARNING & DEEP LEARNING TEST

Instructions:

- Attempt all questions.
- Show all necessary steps for numerical and derivation-based questions.
- Draw diagrams wherever required.
- Write answers clearly and explain your reasoning.

1. Explain the working of Linear Regression and derive the equation (NOT GRADIENT DESCENT) used to calculate the optimal weights using the Mean Squared Error (MSE) cost function.

2. A model has a learning rate of 0.1. If the current weight is 2 and the gradient of the cost function with respect to the weight is 4, calculate the new weight after one gradient descent update. Explain what happens if the learning rate is too large.

3. Why is the Sigmoid function used in Logistic Regression? Explain how its output can be converted into a binary class prediction.

4. Explain Entropy and Information Gain. Using the following dataset, calculate the Information Gain for the features Weather and Windy, and determine which feature should be selected for the first split.

Weather	Windy	Play
Sunny	No	No
Sunny	Yes	No
Overcast	No	Yes
Rain	No	Yes
Rain	Yes	No
Overcast	Yes	Yes
Sunny	No	Yes
Rain	No	Yes

5. What is XGBoost? Explain how it improves the performance of decision trees using boosting.

- 6.** Explain the process of forward propagation in an Artificial Neural Network. Explain the role of weights, bias, activation function, and output.
- 7.** What is backpropagation? Explain how the error is propagated backward through the network and how weights are updated.
- 8.** Explain how CNNs detect features from an image. Describe the roles of convolution, pooling, and fully connected layers.
- 9.** What are word embeddings? Explain how words are converted into numerical vectors and why word embeddings are better than one-hot encoding.
- 10.** Explain the basic idea of Self-Attention in Transformers. What are Query, Key, and Value, and why are they used?